Nirbhay Katiyar

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Software Developer focused on building secure, scalable applications and autonomous AI systems. Experienced in distributed systems, cloud-native backend engineering, and offensive security, with responsible vulnerability disclosures at Spotify, IBM, and LightSpark.

Skills

Languages: Python, Go, C++, SQL, Bash

Backend & APIs: Go (Gin, Fiber), FastAPI, Flask, REST APIs, gRPC, Protocol Buffers, REGEX

AI & Autonomous Systems: Transformers, LLM Integrations, AI Agents, Agentic Workflows, RAG, LLM Orchestration, MCP, Vector DBs

Cloud & DevOps: AWS (Lambda, EC2, CloudWatch, Bedrock, EventBridge, S3, DynamoDB), Docker, Serverless, Microservices, CI/CD

Databases & Messaging: PostgreSQL, Redis, MongoDB, DynamoDB, DragonFly, Pub-Sub

Security: OWASP Top 10, Penetration Testing, JWT Hardening, SQLi, XSS, CSRF, SSRF, CORS, Burp Suite, NMAP, PCAP, Wireshark

CS Fundamentals: Operating Systems, Computer Networking, DBMS, TCP-IP, System Design, OOP, LLD, DSA

Experience

Independent Security Researcher (Bug Bounty) Feb 2024 – Jan 2025
HackerOne *Remote*

- Discovered and responsibly disclosed **critical vulnerabilities** in platforms operated by **Spotify, LightSpark, and IBM**, earning valid reports.
- Researched complex attack chains involving **IDOR, authentication bypass, SSRF, and information disclosure** with clear business impact.

Projects

FlexURL | Production URL Management Platform | *Python, FastAPI, PostgreSQL, AWS* **Live Website**

- Architected a URL management platform using **Twitter Snowflake IDs (Base62)**, supporting 410 redirects/sec with 100% success rate on a 2vCPU server.
- Optimized the redirect pipeline with an asynchronous analytics architecture using ARQ, maintaining a p95 latency below 210ms and an avg of 42ms.
- Engineered advanced features including **custom domains, OS-aware redirection**, password protection, expirations, QR codes, REST APIs, and RBAC.
- Developed a **real-time analytics pipeline** using webhooks, tracking geo-location and device metrics while leveraging Redis to minimize DB load.
- Hardened platform with **SSRF/CSRF mitigation, Google Safe Browsing**, token-bucket rate limiting, Cloudflare edge protection, and centralized logging.

ZeroDaily | Autonomous AI Security News Platform | *Python, Flask, AWS, DynamoDB* **Live Website**

- Built an **autonomous AI pipeline** for scheduled scraping, deduplication, **LLM-powered summarization**, and newsletter delivery with zero manual input.
- Aggregated cybersecurity news daily from **8 security sites and the NVD**, generating AI-curated newsletters for **50+ subscribers**.
- Designed a **fully serverless architecture** (AWS Lambda, EventBridge, DynamoDB, S3), achieving **1.2-second cold starts** and executing in ∼**8 mins**.
- Implemented robust **retry mechanisms** across all pipeline stages and built an administrative dashboard for monitoring subscriber metrics.

TOKELY - A Hybrid Stateless Auth Library | *Python, Argon2, JWT* **PyPI Package**

- Built a **hybrid-state authentication library** in Python supporting **database adapters and JWT** implementations via built-in decorators with over 1000 Users.
- Integrated **Argon2 hashing** and HTTP-only cookies to securely store JWTs, supporting token blacklisting and automated refresh logic.

Open Source Contributions

fastapi-new *(Python)*: Implemented resilient error handling during initialization to detect network failures and clean stale files, resolving persistent crash loops.

fastapi-users *(Python, PostgreSQL)*: Resolved database limit overflow exceptions by refactoring fixed-length hash inputs to dynamic text fields.

Education

Galgotias College of Engineering & Technology Greater Noida, UP
B.Tech in Electronics & Communication Engineering (Expected Apr 2029) *CGPA: 7.59*